

# THE BEST SMART TV WITHIN ₹ 30,000

Smart TVs offer much more than just a good screen and display resolution

**B**uying a television is like welcoming a brand new member to the house. Over the years, the beloved entertainment box has evolved to actually become connected and smart. Keeping that in mind, we look into the essential features to help you choose the best TV within ₹ 30,000.

## Essential features

We all are familiar with TV jargon like display, audio and ports. Added to them now are 'smart' features like connectivity and Internet access. Rohan Agarwal, vice president-consumer electronics business, Micromax Informatics, comments, "Technical parameters that speak about the quality of a smart TV are mainly its panel and internal components. The processing setup of the television should be stable and not crash while installing apps."

Arjuun Bajaj, director, Daiwa TV, explains, "Key technical specifications worth paying attention to include screen size, RAM and ROM, panel grade and brightness, speakers (to give the surround sound) and, most importantly, connectivity. These are the prerequisites to a smart TV experience."

Let us now explore these features for a smart TV in detail.

**Refresh rate, contrast ratio and resolution.** Simply put, refresh rate is the pace at which the TV screen changes images without any motion blur. Technically, it is the number of frames the TV can display every second and is represented in Hertz (Hz). A refresh rate of 120Hz or higher is the optimal specification. A 120Hz refresh rate means the TV can show 120 images per second.

Contrast of the TV is the ratio of the brightest (white) image produced

to the darkest (black) image. There is no proper standard for comparison of contrast ratio as the ratio evaluation process differs from manufacturer to manufacturer. However, a contrast ratio of 3000:1 is advised. Lower contrast ratio results in lower colour clarity or whitewash effect.

Coming to resolution, full HD (FHD) of 1920 × 1080 pixels is a good choice. A 4K resolution, that is 4096 × 2160 pixels, will be good if you can afford a larger screen (over 102cm). Bajaj says, "One should look for picture quality giving a 4K or FHD resolution paired with good audio quality. Considering the sleeker screen size, speaker qualities are being compromised." While at the store, test the TV speakers for audio quality.

**Screen technology.** Till recently, liquid-crystal display (LCD) dominated the market. This type of screen consists of liquid crystal blocks that develop the colour and display the image created with the help of CCFL (cold cathode fluorescent lamp) backlight.

LCD technology has been overtaken by LED screen options. LED TVs available in the market are actually LED-backlit LCD screens. Backlit LED TVs are quite popular among consumers due to their fast refresh rate and long life at an affordable cost. Their picture quality is also quite decent. If you are on a budget, LED TV would make a good option.

Going by specifications, the best option in the market right now is organic light-emitting diode (OLED). However, OLED TVs are costlier too.

**Display panel.** In-panel switching (IPS) panels offer 36-degree viewing angle as against vertical alignment (VA) panels' 20-degree. In terms of colour

Paromik Chakraborty



Micromax Bingbox



Kodak 40FHDXSMART



Mitashi MIDE040v03 FS